

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

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ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA1407	Course Title:	COMPILER DESIGN
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	40% of CIA		
CO1 Statement:	Apply lexical analysis for token specification and recognition using LEX tool. (BL3)		
Student Name:	Ramya.S.S	Reg. No.:	192421226
Section / Batch:	B.Tech - IT	Date:	16/07/2026

Code of Conduct

I Ramya.S.S/192421226 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ramya

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total Score (out of 100)
1	4 / 4	3 / 4	4 / 4	4 / 4	3 / 4	91.25
2	4 / 4	4 / 4	3 / 4	4 / 4	4 / 4	93.75
3	3 / 4	4 / 4	3 / 4	4 / 4	3 / 4	85
4	4 / 4	4 / 4	3 / 4	3 / 4	4 / 4	88.75
5	3 / 4	3 / 4	4 / 4	4 / 4	4 / 4	90
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	89.75
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

N. D. 277

Name: Ramya.S.S

Date: 15.07.2026

Regno: 192421226

Course Code: LSA1407

Course: Compiler Design

Assessment Tool-1 (ANALYTICAL PROBLEM SOLVING)

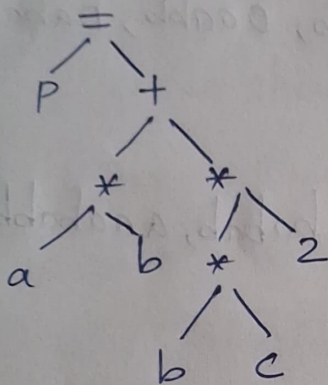
① Compiler Phases for Expression

a) Tokens (Lexical Analysis)

Given: $P = (a * b) + (b * c) * 2$

Tokens generated are $P, =, (, a, *, b,), +, (, b, *, c,), *, 2$

b) Syntax Tree



c) Intermediate Code

$t1 = a * b$

$t2 = b * c$

$t3 = t2 * 2$

$t4 = t1 + t3$

$P = t4$

d) Optimization

$t1 = a * b$

$t3 = t2 * 2$

$t2 = b * c$

$P = t1 + t3$

e) Significance of Target Code

- * Final machine-level instructions
- * Executable by CPU
- * Optimized for performance
- * Uses registers and memory efficiently.

② Regular Expressions

a) $(\epsilon + aa^*)^*$

Valid Strings: $\epsilon, a, aa, aaa, aaaa$

b) $(a^* + b^*)^* abb$

Valid Strings: $abb, aabb, bbbabb, aaabb, bababb$

c) $(a^*b^*)^* aba(a^* + b^*)^*$

Valid Strings: $aba, aaba, abaa, babab, aaababb$

d) $(a+ab)^*$

Valid Strings: $\epsilon, a, ab, aab, abab$

e) $(aba)^* + (a^*b^*)^*$

Valid Strings: $\epsilon, aba, abaaba, aaabbb, ab.$

code:
int a = 10;

char b = "abc";

int c = 30;

char d = "xyz";

in /* comment */ m = 40.5;

Correct Tokens Count: 34

ii) Lexical Errors

Errors:

* in t → should be int

* ch ar → should be char

* 1xab → invalid hexadecimal number

* "abc" → valid, but check Spacing consistency

* Comment inside in /* comment */ breaks keyword.

④ Regular Expressions

i) Even number of 0's

$(1^* 01^* 01^*)^*$

ii) Atleast one 1

$(0+1)^* 1 (0+1)^*$

(iii) Both number of 0's and 1's are odd

$$(0(00)^*1(11)^*)^+$$

iv) Start with 1 and end with 0

$$1(0+1)^*0$$

v) No consecutive 1's

$$(0+10)^*$$

⑤ Given Input String

int var123 = 12345;

float var456 = 123.45;

if (var123 == var456) { return; }

a) Number of Tokens

int, var123, =, 12345, ;, float, var456, =, 123.45, ;, {,

var123, ==, var456, }, {, return, ;, }

Total Tokens = 20

b) Tokens Categorized

1. Keyword

* int

* float

* if

* return

2. Identifiers

* var123

* var456

3. Operators

* =

* ==

4. Seperators

* (

*)

* {

* }

* ;

5. Integer Literal

* 12345

6. Floating-point Literal

* 123.45

c) FSM State Transitions

Count Characters:

* int → 3

* var123 → 6

* = → 1

* 12345 → 5

- * ; $\rightarrow 1$
- * float $\rightarrow 5$
- * var456 $\rightarrow 6$
- * = $\rightarrow 1$
- * 123.45 $\rightarrow 6$
- * ; $\rightarrow 1$
- * if $\rightarrow 2$
- * ($\rightarrow 1$
- * var123 $\rightarrow 6$
- * = $\rightarrow 2$
- * var456 $\rightarrow 6$
- *) $\rightarrow 1$
- * { $\rightarrow 1$
- * return $\rightarrow 6$
- * ; $\rightarrow 1$
- * } $\rightarrow 1$

Total ≈ 66 Transitional

FSM Transitions $\approx 65-70$ transitions

FSM State Transitions

Count Characters:

* int $\rightarrow 3$

* var123 $\rightarrow 6$

* = $\rightarrow 1$

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/for mula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited unclear interpretation or	No interpretation